A counterexample to Oertel's conjecture on Mixed-integer volume

The sharp finite-dimensional form

Abstract

We construct a rational polytope $C \subset \mathbb{R}^{12} \times \mathbb{R}^5$ of positive mixed-integer volume and a number $r < (5/6)^5/2^{12}$ such that every point of $\mathbb{Z}^{12} \times \mathbb{R}^5$ lies in a proper closed halfspace whose intersection with C has at most the fraction r of that volume. The polytope has 4852 nonempty integer fibers. Two selector coordinates and a sign change on four of the other ten integer coordinates produce three families of 252 light fibers. Each heavy fiber paired with a light fiber consists of a simplex and a slab of another simplex sharing a facet. The volume comparisons use exact rational arithmetic. Lean 4 verifies the uniform depth bound.

1 The result

Conjecture 17 of Basu and Oertel [1] asserts that every convex body in $\mathbb{R}^n \times \mathbb{R}^d$ of positive mixed-integer volume has a point in $\mathbb{Z}^n \times \mathbb{R}^d$ of normalized halfspace depth at least $2^{-n}(d/(d+1))^d$. We disprove this bound at $(n, d) = (12, 5)$, where it equals $(5/6)^5/2^{12}$. Cheng and Basu [2] use the name Oertel's conjecture in their Conjecture 2.3 for the lower bound $1/(e2^n)$.

Let vol denote Lebesgue measure on $\mathbb{R}^5$. The *mixed-integer volume*, or *mass*, of a set $D \subseteq \mathbb{R}^{12} \times \mathbb{R}^5$ with measurable slices is $\mu(D) = \sum_{\mathbf{z} \in \mathbb{Z}^{12}} \mathrm{vol}\{\mathbf{x} \in \mathbb{R}^5 : (\mathbf{z}, \mathbf{x}) \in D\}$.

Theorem 1.1. There exist a rational polytope $C \subset \mathbb{R}^{12} \times \mathbb{R}^5$ with nonempty interior, $\mathbf{0} \in C$ and $0 < \mu(C) < \infty$, and a real number r with $0 < r < (5/6)^5/2^{12}$, such that the following holds. For every $(\mathbf{z}, \mathbf{x}) \in \mathbb{Z}^{12} \times \mathbb{R}^5$ there are $b \in \mathbb{R}$ and a nonzero linear functional F on $\mathbb{R}^{12} \times \mathbb{R}^5$ with $F(\mathbf{z}, \mathbf{x}) \geq b$ and $\mu(C \cap \{F \geq b\}) \leq r\, \mu(C)$.

The *depth* of a mixed-integer point is the infimum of $\mu(C \cap H)/\mu(C)$ over proper closed halfspaces H containing it. The theorem bounds the supremum of these depths by r. In units of the fiber over the origin, the construction gives a containing halfspace of mass at most $(5h/6)^5$ at every mixed-integer point, while the total mass W exceeds $4096h^5$. This gives $r = (5h/6)^5/W$.

2 Continuous fibers

Throughout, we fix the rational numbers

$$a = \frac{663}{1000}, \qquad \delta = \frac{664201}{10^6}, \qquad \varepsilon = \frac{13}{20000}, \qquad h = \frac{1111946415}{10^9}, \qquad \eta = \Big(\frac{h}{1+2\delta}\Big)^5,$$

$$(t_0, \dots, t_5) = 10^{-9}(166735371,\ 166735294,\ 166324902,\ 166734811,\ 166734811,\ 166734811).$$

They satisfy $\sum_i t_i = 1$, $1/2 < a < 1$, $0 < \delta < 1$, $\varepsilon < 1/1000$ and $1/7 < t_i < 1/5$. Also $2h < 1 + 2\delta < 3$, which gives $\eta < 1/32 < a/(1+a)$. The only other numerical facts we use are the strict comparisons at the end of Section 5.

Let $B = \{\mathbf{x} \in \mathbb{R}^6 : x_i \geq 0,\ \sum_i x_i = 1\}$ be the standard simplex with vertices $\mathbf{v}_1, \dots, \mathbf{v}_6$. We identify the hyperplane $\sum_i x_i = 1$ with $\mathbb{R}^5$ by dropping x_6 and subtracting $1/6$ from the other coordinates. The centroid of B is then the origin of $\mathbb{R}^5$, scalar multiples of subsets are dilations about the centroid, and $x_1, \dots, x_6$ are affine functions on $\mathbb{R}^5$ with sum 1. From now on vol and μ are measured in units of the Lebesgue volume of B, which does not change the ratios in Theorem 1.1.

For $j = 1, 2, 3$, with these three indices taken cyclically, put $f_j = x_j - a x_{j+1} - (1-a) x_{j+2}$. Each f_j vanishes at the centroid and equals 1 at $\mathbf{v}_j$, so it is a nonzero linear functional on $\mathbb{R}^5$. Moreover $|f_j| \le 1$ on B and $f_1 + f_2 + f_3 = 0$. The cyclic permutation of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ is a volume-preserving linear symmetry of B that carries f_j to f_{j+1}, so we describe the sets below for $j = 1$ and obtain the others by rotation.

Each f_j is a homogeneous linear form in (x_1, x_2, x_3). Non-strict inequalities $f_j \ge 0$ or $f_j \le 0$ therefore cut out a join of their trace on the active triangle $\operatorname{conv}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ with $\operatorname{conv}(\mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6)$. Indeed, normalizing the first three coordinates by their sum when it is positive preserves the inequalities. Points with that sum zero belong to the latter triangle. The simplex spanned by three points of the active triangle and $\mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6$ has volume equal to the absolute determinant of the 3×3 matrix of barycentric coordinates of these points, because the affine map carrying the vertices of B to the vertices of this simplex acts on barycentric coordinates as the direct sum of this matrix and the identity. We call this the determinant formula. On the active triangle, f_1 is positive at $\mathbf{v}_1$, negative at $\mathbf{v}_2$ and $\mathbf{v}_3$, and zero at $(a\mathbf{v}_1 + \mathbf{v}_2)/(1+a)$ and $((1-a)\mathbf{v}_1 + \mathbf{v}_3)/(2-a)$. So $B \cap \{f_1 \ge 0\}$ is a simplex, and the determinant formula gives its volume $\beta = 1/[(1+a)(2-a)]$.

Let $T_j = B \cap \{f_{j+1} \le 0,\ f_{j+2} \le 0\}$, so that $f_j \ge 0$ on T_j and $f_i \le 0$ on T_j for $i \ne j$. Among the vertices of the active triangle, f_2 is positive only at $\mathbf{v}_2$ and f_3 only at $\mathbf{v}_3$. The lines $f_2 = 0$ and $f_3 = 0$ meet at the centroid $\bar{\mathbf{v}} = (\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3)/3$ of the triangle. They meet the edges from $\mathbf{v}_1$ to $\mathbf{v}_2$ and from $\mathbf{v}_1$ to $\mathbf{v}_3$ at $\mathbf{w}_2 = (\mathbf{v}_1 + (1-a)\mathbf{v}_2)/(2-a)$ and $\mathbf{w}_3 = (\mathbf{v}_1 + a\mathbf{v}_3)/(1+a)$, respectively. The edge from $\mathbf{v}_2$ to $\mathbf{v}_3$ has $f_1 < 0$ throughout, so it is disjoint from T_1. The trace of T_1 is the quadrilateral with vertices $\mathbf{v}_1, \mathbf{w}_2, \bar{\mathbf{v}}, \mathbf{w}_3$, whose diagonal from $\mathbf{v}_1$ to $\bar{\mathbf{v}}$ lies on the line $x_2 = x_3$. Consequently $T_1 = U \cup P$, where

$$U = \operatorname{conv}(\mathbf{v}_1, \bar{\mathbf{v}}, \mathbf{w}_3, \mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6), \qquad P = \operatorname{conv}(\mathbf{v}_1, \bar{\mathbf{v}}, \mathbf{w}_2, \mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6).$$

These simplices lie on opposite sides of the hyperplane $x_2 = x_3$ and meet in a common facet, which is a null set. By the determinant formula, U has volume $\theta = a/[3(1+a)]$ and P has volume $\phi = (1-a)/[3(2-a)]$.

To truncate T_1, let $\nu_0, \dots, \nu_5$ be the barycentric coordinates of U in the listed vertex order and put $q_i = \nu_i(\mathbf{w}_2)$. Comparing active coordinates gives $q_1 = 3(1-a)/(2-a)$, $q_2 = -(1-a^2)/[a(2-a)]$, $q_0 = 1 - q_1 - q_2$ and $q_3 = q_4 = q_5 = 0$. Since $1/2 < a < 1$, we have $0 < q_1 < 1$, $q_2 < 0$ and $0 < q_1 + q_2 = q_1(2a-1)/(3a) < 1$, so $0 < q_0 < 1$. Now put $K_1 = T_1 \cap \{\nu_2 \ge \varepsilon q_2\}$. On U we have $\nu_2 \ge 0 > \varepsilon q_2$. On P, let p be the barycentric coordinate at $\mathbf{w}_2$ and, for $i \ne 2$, let p_i be the barycentric coordinate at the vertex of U with index i. Then $\nu_i = p_i + q_i p$ for $i \ne 2$ and $\nu_2 = q_2 p$, since both sides are affine and agree at the vertices of P. Therefore K_1 is the union of U and $P \cap \{p \le \varepsilon\}$. The set $P \cap \{p \ge \varepsilon\}$ is a homothetic copy of P with ratio $1 - \varepsilon$, so $\operatorname{vol}(K_1) = V := \theta + \phi[1 - (1-\varepsilon)^5]$. Rotation gives K_2, K_3 and their coordinates ν_i.

Finally, let $N_j = \operatorname{conv}\big(\eta \mathbf{v}_j + (1-\eta)\mathbf{v}_{j+1}, \mathbf{v}_{j+1}, \mathbf{v}_{j+2}, \mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6\big) \subseteq B$. At these vertices f_j takes the values $\eta - a(1-\eta)$, $-a$, $a-1$, 0, 0, 0, which are nonpositive because $\eta < a/(1+a)$. Hence $f_j \le 0$ on N_j, and $\operatorname{vol}(N_j) = \eta$ by the determinant formula.

3 Integer bases and the polytope

Write $\mathbf{z} = (\mathbf{y}, \mathbf{s}) \in \mathbb{R}^{10} \times \mathbb{R}^2$, with ordinary coordinates $\mathbf{y}$ and selector coordinates $\mathbf{s}$, and put $u = \sum_i y_i$. The linear forms that organize the construction are

$$\tau(\mathbf{z}) = (u - s_2,\ u - s_1,\ u - s_1 - s_2), \qquad g(\mathbf{z}) = \delta\big(\tau_1(\mathbf{z}) + \tau_2(\mathbf{z}) + \tau_3(\mathbf{z})\big) = \delta(3u - 2s_1 - 2s_2).$$

Let $c_i = -1$ for $i \le 4$ and $c_i = 1$ for $i \ge 5$. For $J \subseteq \{1, \dots, 10\}$, let $\mathbf{1}_J$ be the indicator vector of J, and let $\mathbf{b}_J$ have entries c_i on J and 0 off J. With $\mathbf{e}_1, \mathbf{e}_2$ the unit vectors of $\mathbb{R}^2$, the candidates are $(\mathbf{1}_J, \mathbf{0})$ for $J \neq \emptyset$ and $(\mathbf{b}_J, \mathbf{s})$ for $\mathbf{s} \in \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2\}$, the latter with the following values.

Type j	$\mathbf{s}$	$\tau(\mathbf{b}_J, \mathbf{s})$	Sum of entries	Paired	Near
1	$\mathbf{e}_1$	$(u,\ u-1,\ u-1)$	$3u-2$	$u=1$	$u=0$
2	$\mathbf{e}_2$	$(u-1,\ u,\ u-1)$	$3u-2$	$u=1$	$u=0$
3	$\mathbf{e}_1+\mathbf{e}_2$	$(u-1,\ u-1,\ u-2)$	$3u-4$	$u=1$	$u=2$

The sum of the entries of τ is not divisible by 3 in this table and equals $3|J|$ at $(\mathbf{1}_J, \mathbf{0})$, so it is nonzero on every candidate. A *heavy base* is a candidate multiplied by the sign of this sum. After this orientation, the paired value of u in type j gives $\tau_j = 1$ and zeros elsewhere, the near value gives $\tau_j = 0$ and ones elsewhere, and every other value of u gives three positive entries, as do the bases $(\mathbf{1}_J, \mathbf{0})$. We call these heavy bases *paired* of type j, *near* of type j and *regular*, respectively. For each paired base $\mathbf{z}$ of type j, the point $-\mathbf{z}$ is a *light* base of type j. The set $\mathcal{Z}$ of *bases* consists of $\mathbf{0}$ and all heavy and light bases.

Modulo 2, an oriented $(\mathbf{b}_J, \mathbf{s})$ reduces to $(\mathbf{1}_J, \mathbf{s})$, and the three selectors have distinct nonzero residues. Hence the heavy bases represent each nonzero class of $(\mathbb{Z}/2\mathbb{Z})^{12}$ exactly once, and there are 4095 of them. Since $u(\mathbf{b}_J) = |J \triangle \{1,2,3,4\}| - 4$, toggling membership in the first four positions shows that $u = 1$ has $\binom{10}{5} = 252$ solutions in each type. So there are 756 paired bases, 756 light bases and 3339 other heavy bases. The same count at $u = 0$ and $u = 2$ gives $\binom{10}{4} = \binom{10}{6} = 210$ near bases per type, leaving 2709 regular bases. A light base $-\mathbf{z}$ has its partner's residue and differs from it, so the 4852 points of $\mathcal{Z}$ are distinct. We assign to them the following fibers.

Base $\mathbf{z}$	Number	$g(\mathbf{z})$	$\tau(\mathbf{z})$	Fiber $A_{\mathbf{z}}$	$\mathrm{vol}(A_{\mathbf{z}})$
$\mathbf{0}$	1	0	$\mathbf{0}$	B	1
paired, type j	252	δ	$\tau_j = 1$, others 0	$(1+\delta)K_j$	$(1+\delta)^5 V$
light, type j	252	$-\delta$	$\tau_j = -1$, others 0	$(1-\delta)T_j$	$(1-\delta)^5(\theta+\phi)$
near, type j	210	2δ	$\tau_j = 0$, others 1	$(1+2\delta)N_j$	h^5
regular	2709	$\ge 3\delta$	all entries ≥ 1	hB	h^5

The volume of a near fiber is $(1+2\delta)^5\eta = h^5$. Every fiber lies in $(1+g(\mathbf{z}))B$, and the radius $1+g(\mathbf{z})$ is positive because $\delta < 1$. For regular bases this containment uses $h < 1+2\delta < 1+g(\mathbf{z})$ and $\mathbf{0} \in B$. Let $C = \mathrm{conv}\big(\bigcup_{\mathbf{z}\in\mathcal{Z}} \{\mathbf{z}\} \times A_{\mathbf{z}}\big)$. The light fibers of one type have total volume $L = 252(\theta+\phi)(1-\delta)^5$, and all fibers have total volume $W = 3339h^5 + 756(1+\delta)^5 V + 1 + 3L$.

4 The integer fibers of C

We first show that a nonzero integer point of $\mathrm{conv}(\mathcal{Z})$ has only the trivial representation.

Lemma 4.1. If $\mathbf{z} \in \mathbb{Z}^{12} \setminus \{\mathbf{0}\}$ is a convex combination $\sum_a \lambda_a \mathbf{z}^a$ of points $\mathbf{z}^a \in \mathcal{Z}$ with all $\lambda_a > 0$, then $\mathbf{z}^a = \mathbf{z}$ for every a.

Proof. All entries of all bases lie in $\{-1, 0, 1\}$, so $\mathbf{z} = (\mathbf{y}, \mathbf{s})$ has the same property. If a coordinate of $\mathbf{z}$ equals ± 1, or if the contributors $\mathbf{z}^a = (\mathbf{y}^a, \mathbf{s}^a)$ take only two consecutive integer values in that coordinate, then integrality forces all contributors to agree in it. We call this saturation.

Suppose first that $\mathbf{s} \neq \mathbf{0}$, say $s_k = \pm 1$. Saturation gives $s^a_k = s_k$ for all a. Every nonzero selector of a base is $\pm\mathbf{e}_1$, $\pm\mathbf{e}_2$ or $\pm(\mathbf{e}_1 + \mathbf{e}_2)$, so the other selector coordinate of each contributor

lies in $\{0, s_k\}$, and saturation gives $\mathbf{s}^a = \mathbf{s}$. Let $\omega = \pm 1$ be the sign of the nonzero entries of $\mathbf{s}$. By construction, a base with this selector has ordinary part $\omega \mathbf{b}_J$, so $y_i^a \in \{0, \omega c_i\}$ for each i, and saturation gives $\mathbf{y}^a = \mathbf{y}$.

Now suppose that $\mathbf{s} = \mathbf{0}$. Then $y_i = \pm 1$ for some i, and saturation gives $y_i^a = y_i$ for all a. A contributor with a nonzero selector of sign ω has $y_i = \omega c_i$, so $c_i y_i (s_1^a + s_2^a) = \omega(s_1^a + s_2^a) > 0$, while a contributor with selector $\mathbf{0}$ gives 0. The weighted sum of these numbers is $c_i y_i (s_1 + s_2) = 0$, so all contributors have selector $\mathbf{0}$. Their ordinary parts lie in $\{0, 1\}^{10}$, and saturation gives $\mathbf{y}^a = \mathbf{y}$. □

Consequently $\mathbb{Z}^{12} \cap \operatorname{conv}(\mathcal{Z}) = \mathcal{Z}$, and no nonzero base lies in the convex hull of the other bases. For a point $\mathbf{w}$ outside a compact convex set, let $\mathbf{w}_0$ be a nearest point in the set. Minimizing the squared distance to $\mathbf{w}$ on segments from $\mathbf{w}_0$ gives $\langle \mathbf{w} - \mathbf{w}_0, \mathbf{w}' - \mathbf{w}_0 \rangle \le 0$ for every point $\mathbf{w}'$ of the set. The inner product with $\mathbf{w} - \mathbf{w}_0$ is thus strictly larger at $\mathbf{w}$ than on the set. In particular, each nonzero $\mathbf{z} \in \mathcal{Z}$ has a linear functional $\ell_{\mathbf{z}}$ on $\mathbb{R}^{12}$ with $\ell_{\mathbf{z}}(\mathbf{z}') < \ell_{\mathbf{z}}(\mathbf{z})$ for all $\mathbf{z}' \in \mathcal{Z} \setminus \{\mathbf{z}\}$.

Lemma 4.2. The set C is a rational polytope with nonempty interior and $\mathbf{0} \in C$. For $\mathbf{z} \in \mathbb{Z}^{12}$, the slice $\{\mathbf{x} : (\mathbf{z}, \mathbf{x}) \in C\}$ equals $A_{\mathbf{z}}$ if $\mathbf{z} \in \mathcal{Z}$ and is empty otherwise. Consequently $\mu(C) = W$, and every closed halfspace H satisfies $\mu(C \cap H) = \sum_{\mathbf{z} \in \mathcal{Z}} \operatorname{vol}\big(A_{\mathbf{z}} \cap \{\mathbf{x} : (\mathbf{z}, \mathbf{x}) \in H\}\big)$.

Proof. The fibers are polytopes with rational vertices, because T_j and K_j are cut from B by rational inequalities, N_j has rational vertices, and the dilation factors are rational. So C is the convex hull of finitely many rational points. It contains $\{\mathbf{0}\} \times B$, hence the origin. The set B spans the continuous directions. The regular bases $(\mathbf{1}_{\{i\}}, \mathbf{0})$ and the near bases $(\mathbf{0}, -\mathbf{e}_1)$ and $(\mathbf{0}, -\mathbf{e}_2)$ span the integer directions, so C has nonempty interior.

Let $(\mathbf{z}, \mathbf{x}) \in C$ with $\mathbf{z}$ integer, and write $(\mathbf{z}, \mathbf{x}) = \sum_a \lambda_a (\mathbf{z}^a, \mathbf{x}^a)$ with $\lambda_a > 0$, $\sum_a \lambda_a = 1$, $\mathbf{z}^a \in \mathcal{Z}$ and $\mathbf{x}^a \in A_{\mathbf{z}^a}$. If $\mathbf{z} \neq \mathbf{0}$, then Lemma 4.1 gives $\mathbf{z}^a = \mathbf{z}$ for all a, so $\mathbf{z} \in \mathcal{Z}$ and $\mathbf{x} \in A_{\mathbf{z}}$ by convexity of $A_{\mathbf{z}}$. If $\mathbf{z} = \mathbf{0}$, then $\mathbf{x}^a / (1 + g(\mathbf{z}^a)) \in B$ for all a. Since g is linear, the positive weights $\lambda_a (1 + g(\mathbf{z}^a))$ sum to $1 + g(\mathbf{z}) = 1$. Thus $\mathbf{x}$ is a convex combination of the points $\mathbf{x}^a / (1 + g(\mathbf{z}^a))$ of B with these weights, which gives $\mathbf{x} \in B = A_{\mathbf{0}}$. The reverse inclusions are immediate, and the formulas for μ follow by summing over $\mathcal{Z}$. □

5 Halfspaces of small mass

Proof of Theorem 1.1. By Lemma 4.2, a point $(\mathbf{z}, \mathbf{x}) \in \mathbb{Z}^{12} \times \mathbb{R}^5$ either lies outside C or has $\mathbf{z} \in \mathcal{Z}$ and $\mathbf{x} \in A_{\mathbf{z}}$. All inequalities defining the covers below are nonstrict, so points on the boundaries of fibers need no separate treatment.

Exterior points. If $(\mathbf{z}, \mathbf{x}) \notin C$, nearest-point separation gives a nonzero F with $F(\mathbf{z}, \mathbf{x}) > F$ on C. With $b = F(\mathbf{z}, \mathbf{x})$, the halfspace $\{F \ge b\}$ contains the point and misses C, so it has mass 0.

The origin and the light fibers. For $j = 1, 2, 3$, let $H_j = \{(\mathbf{z}, \mathbf{x}) : f_j(\mathbf{x}) \ge 3\tau_j(\mathbf{z})\}$, a proper closed halfspace because $f_j \neq 0$. We read its trace on each fiber from the table in Section 3. Over the origin the trace is $B \cap \{f_j \ge 0\}$, of volume β. If $\tau_j(\mathbf{z}) = -1$, the condition $f_j \ge -3$ holds on the whole light fiber of type j. If $\mathbf{z} \neq \mathbf{0}$ and $\tau_j(\mathbf{z}) = 0$, then $\mathbf{z}$ is a paired or light base of a type $k \neq j$, or a near base of type j. Its fiber lies in $\{f_j \le 0\}$, so the trace lies in the hyperplane $\{f_j = 0\}$, which is null because f_j is nonconstant. If $\tau_j(\mathbf{z}) \ge 1$, the condition requires $f_j \ge 3$, which fails because every heavy fiber lies in $(1 + 2\delta)B$, where $|f_j| \le 1 + 2\delta < 3$. Hence $\mu(C \cap H_j) = \beta + L$. Since $f_1 + f_2 + f_3 = 0$, each $\mathbf{x} \in B$ has $f_j(\mathbf{x}) \ge 0$ for some j, so $(\mathbf{0}, \mathbf{x}) \in H_j$. A point of a light fiber of type j lies in H_j.

Isolated cuts. Let $\mathbf{z} \neq \mathbf{0}$ be a base, let ψ be a nonconstant affine function on $\mathbb{R}^5$, and let $\kappa \in \mathbb{R}$. For $M > 0$ put $F(\mathbf{z}', \mathbf{x}') = \psi(\mathbf{x}') - \psi(\mathbf{0}) + M\ell_{\mathbf{z}}(\mathbf{z}')$ and $b = \kappa - \psi(\mathbf{0}) + M\ell_{\mathbf{z}}(\mathbf{z})$. Then F is a

nonzero linear functional, and $F(\mathbf{z},\mathbf{x}') \geq b$ if and only if $\psi(\mathbf{x}') \geq \kappa$. Since $\mathcal{Z}$ is finite, the fibers are compact and $\ell_{\mathbf{z}}(\mathbf{z}') < \ell_{\mathbf{z}}(\mathbf{z})$ for $\mathbf{z}' \neq \mathbf{z}$, we have $F < b$ on every fiber over $\mathbf{z}' \neq \mathbf{z}$ once M is large. By Lemma 4.2, the isolated cut $\{F \geq b\}$ then has mass $\mathrm{vol}(A_{\mathbf{z}} \cap \{\psi \geq \kappa\})$.

Regular and near fibers. Such a fiber is a simplex of volume h^5. Some barycentric coordinate of $\mathbf{x}$ in it is at least $1/6$. That coordinate is affine and nonconstant, and the cap of the simplex where it is at least $1/6$ is a homothetic copy with ratio $5/6$. So the isolated cut has mass $(5h/6)^5$.

Paired fibers. Let $\mathbf{z}$ be a paired base of type j and let $\mathbf{x} \in (1+\delta)K_j$. By rotation we compute with $j = 1$. Since $\sum_i \nu_i = \sum_i t_i = 1$ on all of $\mathbb{R}^5$, there is an index i with $\nu_i(\mathbf{x}/(1+\delta)) \geq t_i$. The function $\mathbf{x}' \mapsto \nu_i(\mathbf{x}'/(1+\delta))$ is affine and nonconstant, so the isolated cut with $\kappa = t_i$ contains $(\mathbf{z},\mathbf{x})$ and has mass $(1+\delta)^5 V_i$, where $V_i = \mathrm{vol}(K_1 \cap \{\nu_i \geq t_i\})$.

To compute V_i, we split K_1 into U and $P \cap \{p \leq \varepsilon\}$, which overlap in a null set. In U the cap $\{\nu_i \geq t_i\}$ is a homothetic copy of U with ratio $1 - t_i$, of volume $\theta(1-t_i)^5$. On P with $i = 2$ we have $\nu_2 = q_2 p \leq 0 < t_2$, so this part is empty. Let $i \neq 2$. For $0 \leq p \leq \varepsilon$, the bounds on q_i and t_i give $t_i - q_i p > 0$, $1 - t_i - (1-q_i)p > 0$ and $1 - q_i > 0$. In the coordinates p and p_k with $k \neq i, 2$, the simplex P is a standard simplex of Lebesgue volume $1/120$, and its volume in our units is ϕ. Substituting $p_i = 1 - p - \sum_{k \neq i,2} p_k$ turns the condition $\nu_i = p_i + q_i p \geq t_i$ into $\sum_{k \neq i,2} p_k \leq 1 - t_i - (1-q_i)p$, and it forces $p_i \geq t_i - q_i p > 0$. So for each p the four remaining coordinates range over a simplex of Lebesgue volume $[1 - t_i - (1-q_i)p]^4/24$, and the contribution of P is

$$5\phi \int_0^{\varepsilon} \left[1 - t_i - (1-q_i)p\right]^4 dp = \phi \, \frac{(1-t_i)^5 - \left(1 - t_i - (1-q_i)\varepsilon\right)^5}{1 - q_i}.$$

Thus V_i is $\theta(1-t_i)^5$ plus this quantity for $i \neq 2$, and $V_2 = \theta(1-t_2)^5$.

Exact comparisons. The cuts above reduce the theorem to comparisons of explicit rational numbers. In lowest terms, $\theta = 221/1663$, $\phi = 337/4011$, $\beta = 1000000/2223431$, $q_0 = 776569/886431$, $q_1 = 1011/1337$, $q_2 = -560431/886431$ and $5h/6 = 74129761/80000000$. Exact evaluation of the formulas above gives the following floors, and the checker in Appendix A repeats these comparisons with exact fractions.

Margin	$\lfloor 10^{12} \cdot \mathrm{margin} \rfloor$
$(5h/6)^5 - \beta - L$	1262439
$(5h/6)^5 - (1+\delta)^5 V_i$ for $i = 0, \ldots, 5$	7542, 6288, 1211, 7869, 7869, 7869
$W - 4096h^5$	45459473

All floors are positive, so $\beta + L < (5h/6)^5$, $(1+\delta)^5 V_i < (5h/6)^5$ for $i = 0, \ldots, 5$, and $4096h^5 < W$. Therefore every point of $\mathbb{Z}^{12} \times \mathbb{R}^5$ lies in a proper closed halfspace H with $\mu(C \cap H) \leq (5h/6)^5$. Put $r = (5/6)^5 h^5 / W$. Since $\mu(C) = W$, we get $\mu(C \cap H) \leq r\,\mu(C)$. Since $W > 4096h^5$, we also get $0 < r < (5/6)^5/2^{12}$. Numerically, $W = 6962.7250497\ldots$, $10^5 r = 9.811464096\ldots$ and $10^5 (5/6)^5/2^{12} = 9.811464160\ldots$. Lemma 4.2 gives the remaining properties of C. □

References

[1] Amitabh Basu and Timm Oertel. *Centerpoints: A link between optimization and convex geometry*. Manuscript version with Conjecture 17.

[2] Hongyu Cheng and Amitabh Basu. *Linear Threshold for Oertel's Conjecture on the Mixed-Integer Volume*. Conjecture 2.3.

A Lean statement and exact replay

The exact checker and Lean project are tracked in PR #14. The verification uses the following pins.

Verification source	`d368306f28ce808d2b2b773cc025745d835538bf`
Lean toolchain	`leanprover/lean4:v4.34.1`
mathlib revision	`d13f23b723b8a846827a245b89c10fc7d3f11612`

In the namespace `SharpOC.Selector17`, the final declaration reads as follows.

```
theorem sharp_oc_counterexample :
    ∃ C : Set Ambient, IsCompact C ∧ Convex ℝ C ∧ (interior C).Nonempty ∧
      ((0 : IntegerSpace), (0 : E)) ∈ C ∧
      0 < mixedIntegerMass C ∧ mixedIntegerMass C ≠ ⊤ ∧
      ∃ r : ℝ, 0 < r ∧ r < (((5 : ℝ) / 6)^5 / (2 : ℝ)^12) ∧
        ∀ (z : Coordinate → ℤ) (x : E),
          ∃ F : Ambient →L[ℝ] ℝ, ∃ b : ℝ, F ≠ 0 ∧ b ≤ F ((fun j => (z j : ℝ)), x) ∧
            (mixedIntegerMass (C ∩ {p | b ≤ F p})).toReal /
              (mixedIntegerMass C).toReal ≤ r
```

Here `Coordinate` is `Fin 10 ⊕ Fin 2`, `IntegerSpace` is `Coordinate → ℝ`, `E` is `Fin 5 → ℝ` and `Ambient` is `IntegerSpace × E`. For a set `D`, the value `mixedIntegerMass D` is the extended nonnegative sum, over all `w : Coordinate → ℤ`, of the Lebesgue volume of `{x | ((fun k => (w k : ℝ)), x) ∈ D}`, which is $\mu(D)$ in the Lebesgue units of Section 1. Finiteness of the mass makes both `toReal` values faithful. Lemma 4.2 establishes rationality of the polytope. With Python 3, Git and elan installed, run the following from the repository root. Lake reads its dependency pins from `lean/lake-manifest.json`.

```
set -eu
python3 -B research/sharp-dimension-17/check.py
cd lean
lake exe cache get
lake build
lake env lean CheckAxioms.lean
lake env lean --stdin <<'LEAN'
import SharpOC.Selector17Counterexample

#check SharpOC.Selector17.integer_projection_census
#check SharpOC.Selector17.ambient_dimension
#check SharpOC.Selector17.sharp_oc_counterexample
#print axioms SharpOC.Selector17.sharp_oc_counterexample
LEAN
```

The checker uses exact rational arithmetic to enumerate the 4852 listed bases and check the strict comparisons. The geometric proof and Lean establish that the list exhausts the integer projection. The Lean build passed, and the dimension lemma and final theorem depend only on `propext`, `Classical.choice` and `Quot.sound`.